Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

System mass

Meaning in my words How would you explain system mass to a friend?	**How can you tell?** If a cart and hanging mass accelerate together, which masses belong in the total?
Example or sketch A loaded cart is 0.8 kg and its loaded hanger is 0.2 kg. What mass accelerates?	**Non-example and why** Why leave the fixed track out of the moving mass?

Control

Meaning in my words How would you explain control to a friend?	**How can you tell?** You test different cart masses. Why use the same hanging weight each time?
Example or sketch Name two things you would set the same before each release.	**Non-example and why** A partner lists cart mass as a control while adding mass each trial. Where does it belong?

Frayer models

SECTION A / VOCABULARY Name _____

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Inverse relationship

Meaning in my words How would you explain inverse relationship to a friend?	**How can you tell?** Keep the net force fixed. If the moving mass doubles, what happens to acceleration?
Example or sketch At 4 N, compare the accelerations of carts of 1 kg and 2 kg.	**Non-example and why** If both mass and net force double, must acceleration halve?

Slope

Meaning in my words How would you explain slope to a friend?	**How can you tell?** At the same net force, which graph should be straight: a against M or a against 1/M?
Example or sketch Acceleration rises by 2 m/s² when 1/M rises by 0.5 per kg. What is the slope?	**Non-example and why** Someone says this slope is the mass. How can you check?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a versus 1/M. Label the axes and units, draw the rising line, and name the quantity represented by its slope.



My marks mean

On assignment: Inverse relationship

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Test engineer**

Build a two-row test card. Pick the model with the larger velocity change and show the mass comparison for the test crew.

☐ **Lab correspondent**

Explain to viewers why two equal pushes can look different. Your producer must check what stayed the same in both trials.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Use mass × acceleration as an independent check.